

Mach-Zehnder Interferometer Design Proposal

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I. INTRODUCTION

Photonics integration has emerged as one of the possible solutions to meet the growing demand for higher speed and bandwidth of data transfer. In response, some foundries have developed the capability to fabricate photonic components. In this report we propose several Mach-Zehnder Interferometer (MZI) designs to characterize the performance of foundry-produced photonic components. By making controlled variations in the design and comparing simulation data with measurement result, we expect to assess the current capabilities and limitations of photonic integrated circuit manufacturing.

II. THEORY

MZI works using the concept of interference. An input light is split using a y-branch to two different paths. The intensity of each split and their electric field is defined by equation 1 and 2.

$$I_{on} = I_i/2 \quad (1)$$

$$E_{on} = (E_i/\sqrt{2}) e^{(-i\beta_n - \frac{\alpha_n}{2})L_n} \quad (2)$$

With β_n representing the propagation constant of light $n \in \{1, 2\}$ and α_n as its loss per unit length.

Afterward, the lights are combined via a reversed y-branch (a combiner) so that the resulting output is a superposition of two branched lights with variability introduced using the different paths. The resulting output of the MZI is approximated by equation 3.

$$E_o = (E_{o1} + E_{o2})/\sqrt{2} \quad (3)$$

With identical waveguide, we can simplify the equation by setting $\beta_n = \beta$ and $\alpha_n = \alpha$ to get output intensity as follows:

$$I_o = \frac{I_i}{4} \left| e^{-i\beta L_1 - \frac{\alpha}{2}L_1} + e^{i\beta L_2 - \frac{\alpha}{2}L_2} \right|^2 \quad (4)$$

In the ideal case, this process should not introduce loss. However, imperfections commonly occurred during fabrication process. This led to distortion and losses from the supposed output signal. The quality of fabrication can be

quantified by comparing simulation result and measured data.

III. MODELLING AND SIMULATION

A. Waveguide

In our simulation, we use waveguide with width 500 nm and height 220 nm. Simulation is done using Lumerical MODE, with wavelength of 1.55 μm . The result of the simulation is shown in Figure 1 (TE) and Figure 2 (TM).

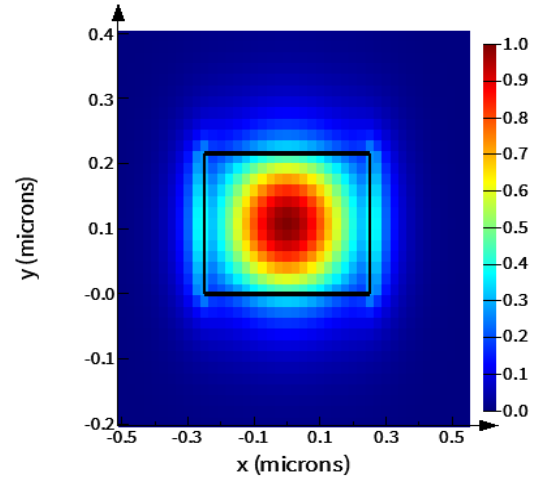


Fig. 1. TE Mode Electric Field Intensity

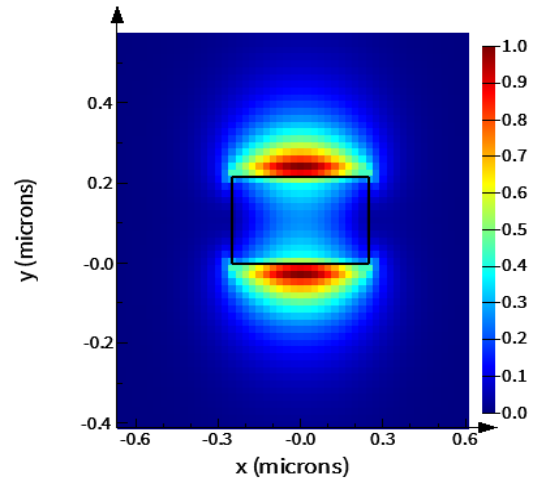


Fig. 2. TM Mode Electric Field Intensity

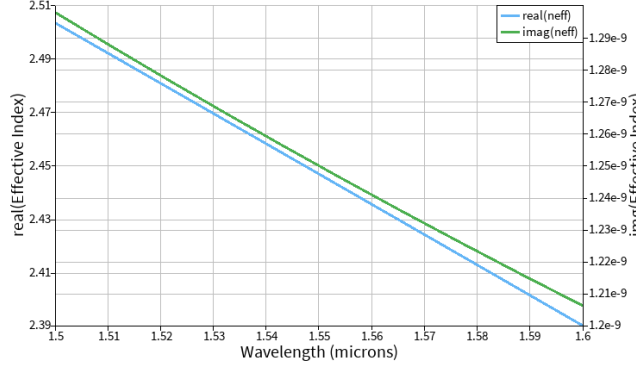


Fig. 3. TE Mode Effective Index

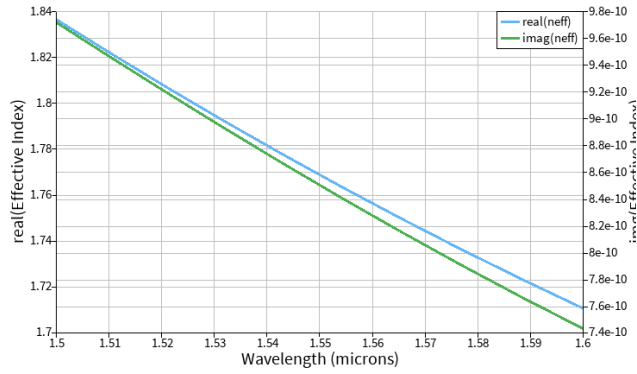


Fig. 4. TM Mode Effective Index

As can be seen from figure 3 and figure 4, effective index decreases as wavelength increase. Effective index (n_{eff}) is around 2.447 for the tracked TE mode and 1.769 for the tracked TM mode at 1.55 μm wavelength.

We use a script to further analyze the simulation data and find the equation for the effective index

$$n_{eff}(\lambda) = 2.46 - 1.13(\lambda - 1.55) - 0.04(\lambda - 1.55)^2 \quad (5)$$

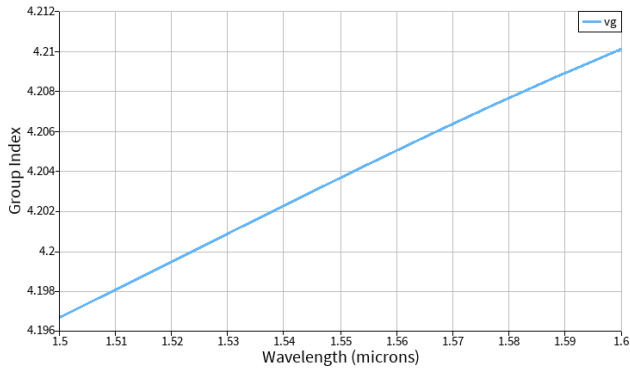


Fig. 5. TE Mode Group Index

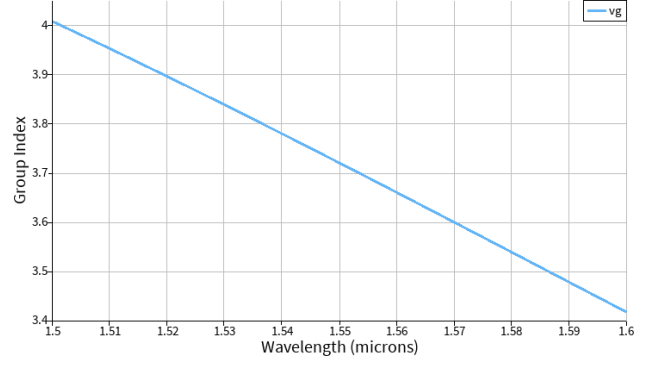


Fig. 6. TM Mode Group Index

The group index however, shows a different trend. Where it increases along with wavelength in TE mode and decreases in the tracked TM mode. This can be seen in figure 5 and figure 6. At 1.55 μm , group index (n_g) for the TE mode is 4.197 and for the TM mode 4.008

B. Mach-Zehnder Interferometer

Our MZI is designed as shown in figure 7. We will vary the length of the upper waveguide strip to introduce phase difference.

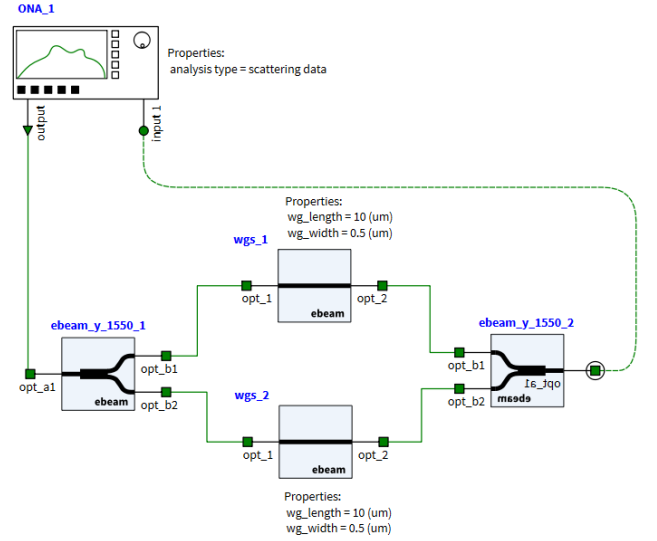


Fig 7. MZI design in ANSYS Interconnect

From equation 4, if we assume the loss is small, we can derive a simpler equation for the MZI transfer function.

$$T_{MZI}(\lambda) = \frac{I_o}{I_i} \quad (6)$$

$$T_{MZI}(\lambda) = \frac{1}{2}(1 + \cos \beta \Delta L) \quad (7)$$

As reference, we listed the output intensity based on simulation result in table 1.

Table 1. Length Difference vs Gain

$\Delta L(\mu m)$	$G(dB)$
50.086	-0.293172
75.762	-27.2257
99.696	-3.77786
125.374	-2.71658

IV. FABRICATION

We designed 5 MZI with different waveguide length difference, each with one copy to take into account the consistency of the fabrication process. The design is as follows:

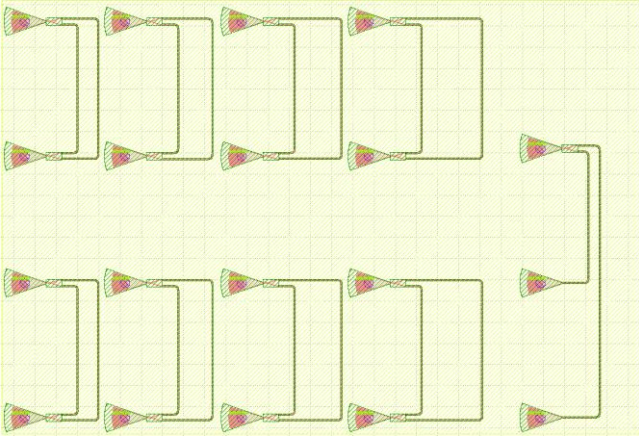


Fig. 9 MZI design in KLayout

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication.

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acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

V. EXPERIMENT DATA

After the fabrication process is completed, the devices were measured using an automated test setup with automated control software written in Python/Matlab [2]. An Agilent tunable laser was used as the input source and Agilent optical power sensors as the output detectors. The wavelength was swept at most from within 1460-1650 or smaller, so that peak wavelength is approximately 1550 nm.

The measurement data gathered from the automated tests are as follows:

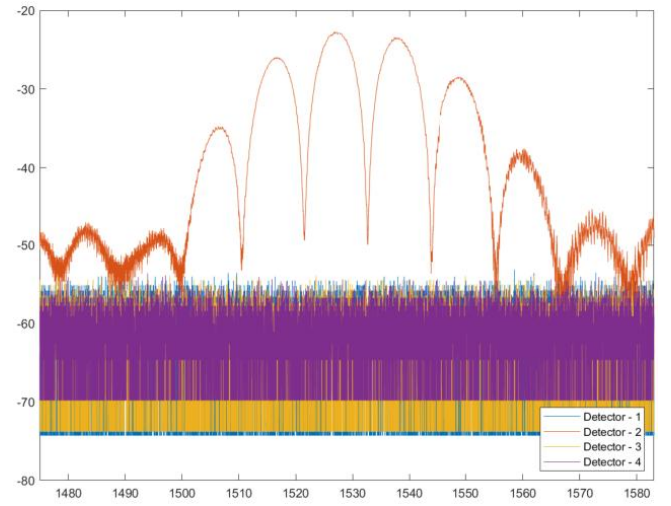


Fig. 10 MZI measurement data for $\Delta L = 50.086 \mu m$

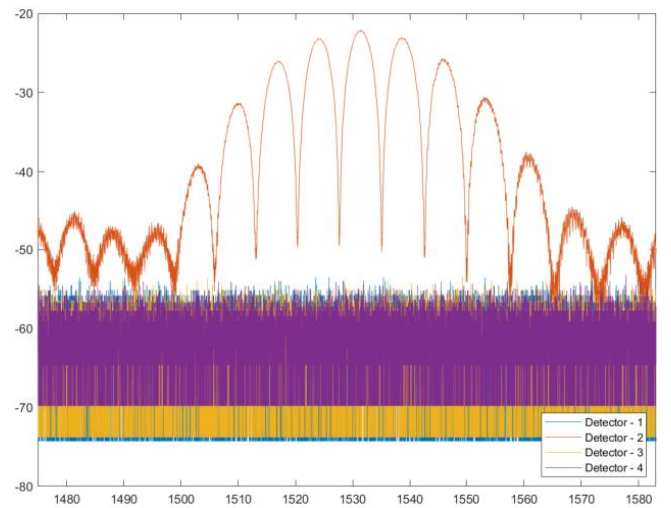


Fig. 11 MZI measurement data for $\Delta L = 75.762 \mu m$

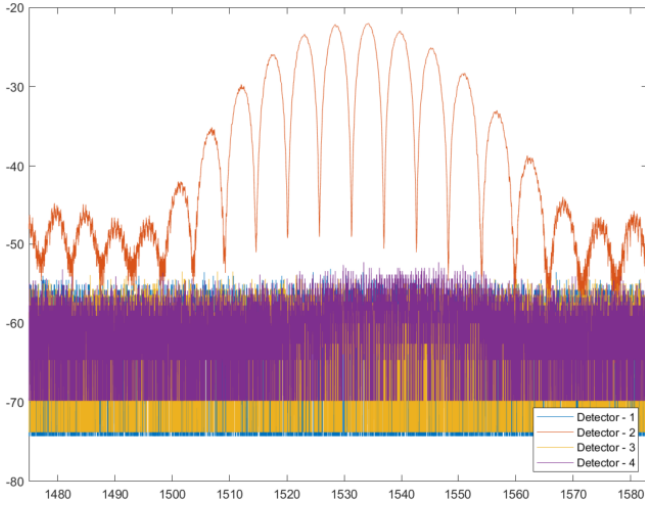


Fig. 12 MZI measurement data for $\Delta L = 99.696 \mu\text{m}$

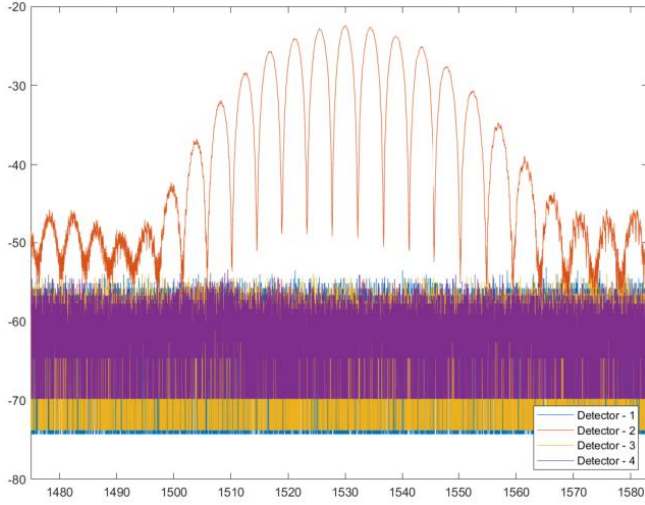


Fig. 13 MZI measurement data for $\Delta L = 125.374 \mu\text{m}$

VI. ANALYSIS

Experimental data is processed by performing baseline correction via polynomial fitting or order 4. Offset is also added to ensure the highest peak is set at 0 dB. The signal gain is the observed at 1550 nm and compared with simulation result. Data acquired is as follows.

Table 2. Simulation Gain and Experimental Gain

$\Delta L(\mu\text{m})$	$G \text{ simulation (dB)}$	$G \text{ measured and processed (dB)}$
50.086	-0.293172	-0.688503
75.762	-27.2257	-24.8508
99.696	-3.77786	-2.42076
125.374	-2.71658	-14.8503

Comparing simulation and measured result, we found that generally the gain at 1550 nm is relatively similar, with the exception at 125.374 nm.

This means that the simulation and experiment data is generally consistent and within a similar phase. The difference at 125.374 nm however, present an opposing conclusion to this statement. This error may come from various sources such as simulation inaccuracy, loss of information from data processing method and manufacturing defect.

VII. CONCLUSION

In this report we designed a basic Mach-Zehnder Interferometer (MZI) with various path length difference and evaluate its gains to simulation result. This report describes the steps for simulating MZI and compares it to real fabricated MZI. We tried various path length difference and refer to gain at a fixed wavelength to compare the similarity. We found that the experiment result generally agrees with the simulation. Validation however must be done to ensure fabrication product performance as high deviation is possible.

REFERENCES

L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge University Press, 2015.